

第4回数理構造セミナー

日時：2018年4月12日（木） 11:00～18:00

場所：すずかけ台キャンパス、S1棟

プログラム：

11:00-11:30 (S1-203) Hayato Goto: “Modeling Evolution of Business Network and Their Interaction”

11:30-14:00 (S1-second floor) Poster session with lunch (S1-203)

14:00-15:15 (S1-317-2) Murat Erkurt: “Emergence of Form in Embryogenesis”

15:30-16:45 (S1-317-2) Mauricio Barahona: “Graph partitioning and community detection based on dynamics”

16:45-18:00 (S1-second floor) Poster session with beer (S1-203)

Posters (to be posted until 20th):

●A. Matsuo: “Power-Law Distributions from Sigma-Pi Structure of Sums of Random Multiplicative Processes”

●E. Takagi: “Category classification of Information diffusion in Blogosphere network based on degree of similarity between documents”

●G. Sakoda: “Parameter estimation method for non-stationary Poisson time series with fluctuation scaling and incomplete observation”

●T. Hamano: “Dealer model of foreign exchange market with finite assets”

●T. Sueshige: “Ecology of Forex market: Human vs. Artificial Intelligence”

●Y. Kobayashi: “Evolutionary Flow and Scaling Laws of Business Firms in 3-dimensional State Space”

●J.-F. Boilard: “Market Microstructure Analysis of Limit Orders in FX Market”

●Y. Shida: “Deriving mass human mobility patterns from individuals location data”

●K. Kanazawa: “Boltzmann Equation for Financial Brownian Motion”

●J. Ozaki: TBA

●F. Rosas: TBA

●T. Nozawa, M. Kondo, R. Yamamoto, H. Jeong, S. Ikeda, K. Sakaki, Y. Ishikawa, Y. Miyake, R. Kawashima: “Shared subjective experience and interpersonal neural synchronization in foreign language active learning classroom: a pilot study”

●K. Ogawa, Y. Miyake: “Robust patterning of gene expression based on internal coordinate system of cells”

- H. Ora, Y. Miyake: “A method for prediction of behavioral errors from single-trial electrophysiological data”
- S. Amano, K. Ogawa, Y. Miyake: “Node property of weighted networks considering connectability to nodes within two degrees of separation”
- T. Mukai, T. Terano: “Effects of Trade Environment in the Decentralized Inter-firm trade Structure Model through Agent Simulation”

Abstract

Murat Erkurt: Development of form in an embryo is the result of a series of symmetry breakings starting from topological and informational singularity of the zygote. Pattern generating reaction-diffusion systems are not capable of form generation, as they are locally isotropic. We introduce Vector-Reaction-Diffusion-Drift system based on three new concepts: source-field coupling, local drift for global symmetry breaking and vector morphogens as pillars of form. An integrated model with topological, physical, chemical and regulatory layers is presented. Simulation of blastula moving into gastrulation shows realistic form generation as an emergent phenomenon of the ensemble of locally interacting identical cells. We further expand our model by introducing Dynamical Cell Differentiation where the regulatory genome is generalized into a finite-state-machine. This allows symmetry breakings to be choreographed into sub-spatial domains resulting in further detailed form generation.

第 5 回数理構造セミナー

日時：2018 年 4 月 19 日（木）14:00～17:30

場所：すずかけ台キャンパス、S1 棟 3 階 317-2 号室

プログラム：

1. 14:00-15:30 Henrik J. Jensen

Title: “Analysis of EEG response from musicians and audience during performances: improvisation versus non-improvisation”

2. 15:45-16:45 Tim Evans

Title: “Networks and Space-time”

Abstract: There are many data sets where the nodes in the most obvious representation come with a natural order so these are Directed Acyclic Graphs. This order can be interpreted as an arrow of time suggesting that the most natural geometrical setting are spacetimes with their non-Riemannian geometry. In this talk I discussed this idea in terms of simple models and analysis of several citation networks.

3. 16:45-17:30 Paul Expert

Title: “Spectral signature of artificial and real dynamical systems”

Abstract: Temporal Graph Signal Analysis is a recent addition to the network analysis toolbox that naturally couples the activity on a network with its structure. In the is talk, I will introduce TGST, and then show how it can be used to characterize different states accessible to the XY model and differentiate between brain states.